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Preliminary analysis of the July 30, 2024, Wayanad landslide disaster in India: Causes and impacts

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ABSTRACT

In recent years, with the increasingly pronounced trend of global climate warming, extreme weather events have become more frequent and intense, leading to a significant rise in the occurrence of natural disasters. Landslides, as a common and highly destructive type of geological hazard, are often extremely damaging when triggered by short-duration intense rainfall. At 2:00 a.m. on July 30, 2024, a large-scale landslide occurred in the Wayanad district of Kerala, India, induced by continuous heavy rainfall. This landslide disaster resulted in at least 373 deaths, over 200 injuries, and 218 people reported missing, making it the most devastating disaster in Kerala's history in terms of both casualties and damage, and causing severe human and economic losses. This study utilizes multi-source data and remote sensing imagery to review the evolution of the disaster event in detail and conducts a preliminary analysis of the landslide's causes based on Disaster System Theory. The results indicate that this large-scale landslide was a reactivation of a pre-existing failure. While extreme tropical rainfall was the primary triggering factor, the geological setting and anthropogenic activities further intensified the disaster's occurrence and impacts. Specifically, the synergistic coupling among the disaster-breeding environment (unstable geological structures and anthropogenic disturbances), disaster-inducing factors (intensified rainfall under climate change), and vulnerability of exposed elements (high population density, insufficient emergency response, and generally high societal risk) collectively led to the disaster. Despite active emergency response efforts following the event, the management process still revealed several shortcomings, including inaccurate early warnings, inadequate in-disaster coordination, and insufficient post-disaster recovery. This study summarizes the experience and lessons from India's response to the rainfall-induced landslide, and, in combination with China's context, proposes targeted countermeasures, aiming to provide scientific evidence and practical guidance for improving emergency management and disaster risk reduction capacities in China and other countries facing similar hazards.

1. Introduction

Landslides are among the most destructive and catastrophic natural hazards worldwide (Guo et al., 2023; Petley, 2012; Huang et al., 2022). They are often accompanied by secondary disasters such as debris flows and rockfalls, and are characterized by a sudden onset and great destructive force (Huang et al., 2025; Shao et al., 2024). In recent years, as global climate change has intensified, extreme rainfall events have become more frequent. Heavy rainfall can saturate soil and rock masses and raise pore water pressure, significantly increasing the risk of landslides (Emberson et al., 2021, 2022; Hakro and Harahap, 2015; Kirschbaum et al., 2020; Ma et al., 2023; Liu and Xu, 2024). Globally, rainfall-induced landslides have occurred with increasing frequency, emerging as one of the most significant geohazards faced by many countries. Numerous catastrophic landslide events in history have been directly associated with episodes of extreme precipitation. For instance, in 1999, the Vargas State of Venezuela experienced exceptionally intense rainfall, which triggered widespread landslides and debris flows, resulting in tens of thousands of fatalities (Larsen et al.). In 2006, persistent heavy rainfall on Leyte Island in the Philippines led to a deep-seated

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landslide that buried an entire village and caused the deaths of over a thousand people (Evans et al., 2007). Similarly, in 2016, southwestern Sri Lanka was affected by continuous torrential rains that triggered multiple landslides, leaving more than 200 people dead and displacing hundreds of thousands, marking one of the most severe landslide disasters in the country's recent history (Dang et al., 2019; Tan et al., 2020).

Among regions worldwide that are frequently affected by landslides, Asia is one of the most severely impacted due to its complex topography and monsoon-driven climatic conditions. Within Asia, India and China are the two countries most frequently affected by landslides and suffer the greatest overall impacts.

In China, due to the influence of complex climatic conditions and diverse topography, landslide hazards triggered by extreme rainfall are especially pronounced. According to the China Statistical Yearbook 2023, between 2000 and 2022 China experienced an average of 12932 landslides per year, accounting for 66 % of all geological disasters nationwide. Excluding seismic triggers, approximately 90 % of landslides in China are attributed to rainfall-induced slope failures (Li, 2005). This is particularly evident during the summer monsoon season, when abundant rainfall leads to frequent shallow landslides across many regions. For example, heavy rainfall triggered landslides in Tianshui, Gansu Province in 2013 (Shao et al., 2023; Huang et al., 2021; Zhang et al., 2023a), Longchuan, Guangdong Province in 2019 (Ma et al., 2022), and Mazhe Village in Enshi, Hubei Province in 2020 (Xue et al., 2022). Furthermore, the record-breaking rainstorm in Henan Province in 2021 (Wei et al., 2023; Xue, 2023; Yin et al., 2022; Wang et al., 2024) and the impacts of Typhoon Doksuri in 2023 caused numerous landslides in the Beijing–Tianjin–Hebei region (Gao et al., 2024; Ma and Wang, 2024; Song et al., 2023; Xiao, 2023; Zhang et al., 2023b; Ma et al., 2025; Xie et al., 2025). The frequent occurrence of such disasters underscores the severity of the rainfall-induced landslide situation, highlighting the need to further strengthen disaster prevention and mitigation efforts in China.

India is likewise among the regions most prone to extreme rainfall-induced landslide disasters. Owing to its geographic and climatic setting, the annual Southwest Monsoon season brings sustained heavy rainfall in India. These intense rains, especially in steep mountainous areas such as the Himalayas and the Western Ghats, readily trigger landslides and debris flows. Historical records show that rainstorm-induced landslides occur frequently in India and cause severe damage. For instance, in mid-June 2013 the state of Uttarakhand was struck by an extreme rainstorm that triggered landslides and flash floods, resulting in over 6000 fatalities. This event ranks among modern India's worst natural disasters, underscoring the immense destructive power of extreme rainfall in triggering cascading hazards in mountainous regions (Bhambri et al., 2017; Gupta et al., 2016; Martha et al., 2015). Only a year later, in July 2014, a massive landslide struck Malin village in Maharashtra during monsoon rains, burying the entire village and killing at least 151 people (Patil and Gopale, 2018; Singh et al., 2016). In August 2018, the southern state of Kerala was hit by the most intense rainstorm in decades, causing severe flooding and triggering thousands of landslides that resulted in 483 deaths across the state. In the Western Ghats, local daily rainfall exceeded the long-term average by more than 36 %, causing numerous slopes to fail in a short time; in total, this event triggered an estimated 4728 landslides (Martha et al., 2019; Arunkumar et al., 2024; Hao et al., 2022).

In recent years, extreme rainfall-triggered landslide disasters in India have shown no signs of abating. On July 30, 2024, following prolonged heavy monsoon rainfall, a large-scale landslide struck the Wayanad region of Kerala, India. This sudden and extremely destructive landslide destroyed roads, houses, and farmland, severely damaging local infrastructure and the environment. According to statistics, it caused at least 373 deaths, over 200 injuries, and left 218 people missing. In the aftermath, the local government promptly launched an emergency response, mobilizing rescue forces to carry out relief operations. This disaster revealed the vulnerability of tropical mountainous ecosystems to extreme weather and highlighted the formidable challenges posed by the coupling of climate change and geological hazards. It also provided valuable lessons for the prevention and emergency response of similar disasters worldwide. Against this backdrop, the present study draws on disaster systems theory to conduct a preliminary analysis of the occurrence process and causative mechanisms of the Wayanad landslide, and further examines the shortcomings in the emergency response to this event, with the aim of providing useful insights for strengthening the emergency management of rainfall-induced landslide disasters in China.

2. Regional overview

Kerala, located in the southwestern part of the Indian Peninsula, is bordered by the Arabian Sea to the west and the Western Ghats to the east. The state has a narrow shape with a wider central region. The terrain varies significantly from west to east, consisting of coastal plains primarily composed of alluvial deposits, a central region covered with red soils, and highland areas characterized by a predominance of Precambrian rocks (Hao et al., 2022). The highest elevation in Kerala reaches 2695 m. Due to its rugged topography, Kerala is highly prone to natural disasters such as landslides. Previous studies have indicated that the state is part of a landslide-prone region, with numerous occurrences of such disasters (Vasudevan and Ramanathan; Sreekumar, 2009). Notable landslide events before the 2018 floods include the Mundakkai landslide in 1984, which resulted in 14 fatalities; the Kappikalam landslide in 1992, causing 11 deaths; and the Vallamthode landslide in 2007, which led to 4 deaths. Other significant landslides include the Amboori landslide in 2001, which claimed 39 lives, the Pamba landslide in 1999, which resulted in 25 deaths, and the Nittumkoottamala landslide in 2004, which caused 10 fatalities (Kuriakose et al., 2009). Since the catastrophic floods of 2018, the frequency and severity of landslides in Kerala have increased significantly, as shown in Table 1.

This rainfall-induced landslide occurred in the Wayanad region of northeastern Kerala, India (Fig. 1a). The area's topography is relatively subdued, with local elevations ranging from approximately 700 to 2100 m above sea level, giving rise to a diverse landscape of hills, valleys, and grasslands (Das, 2024). Geologically, Kerala lies within the Indian Peninsular Craton, one of the major geotectonic units of the Indian subcontinent. Lithologically, the Wayanad region features a variety of rock types belonging to the Peninsular Gneissic Complex, Charnockite Group, Wayanad Group, and Migmatite Complex (Achu et al., 2025; Ambili and Narayana, 2014). Owing to the region's complex geological history and frequent tectonic activity, the bedrock is heavily fractured with well-developed joints and fissures, resulting in relatively poor stability. In addition, these rocks readily weather into loose, unconsolidated materials that become easily saturated during intense rainfall, providing the primary source material for landslides and debris flows. Climatically, it experiences a tropical rainforest climate that is hot and humid with high annual precipitation. Such conditions significantly influence local geological processes and the hydrological cycle, leading to frequent landslides and debris flows in mountainous and hilly areas. Furthermore, Wayanad is known for its tea and spice plantations, which are often situated on steep slopes; this land-use pattern, coupled with the rugged terrain and dense vegetation, also contributes to the frequent occurrence of landslides in the area.

To further investigate the historical landslide occurrences in the region, we examined multiple data sources, including the international Emergency Disaster Database (EM-DAT), the Global Landslide Catalogue (GLC) (Kirschbaum et al., 2010, 2015), the Global Fatal Landslide Database (GFLD) (Froude and Petley, 2018), and historical landslide point data provided by the Geological Survey of India. The spatiotemporal distribution characteristics of historical landslide disasters in the Wayanad region were analyzed (Fig. 1b). As shown in the figure, the southern and southwestern parts of Wayanad are particularly susceptible to landslides. The influence of geological and hydrological processes in these areas increases the likelihood of landslides, particularly in the high-altitude areas of the southwestern region.

Table 1
Historical landslide casualty and disaster records in Kerala, India.

\	Time	Event Description	Source of Information
1	2018	Kerala, in southern India, was severely affected by the most extreme monsoon rainfall since 1924. From June 1 to August 28, the total rainfall in Kerala was 42 % above the annual average, triggering 4728 landslides across the state. Over 1200 villages were impacted, resulting in over 440 deaths and significant property damage. Aside from the relatively flat and flood-prone Alappuzha district, 13 other districts experienced landslide events, with Idukki being the most severely affected.	(Hao et al., 2020, 2022)
2	2019	According to Indian media reports, by August 13, 2019, ongoing heavy rainfall in several regions of India had triggered floods, landslides, and mudslides, resulting in at least 200 fatalities and numerous missing persons. Kerala was particularly hard-hit, with at least 85 fatalities. Additionally, more than 50 people were reported missing in the landslide incidents in Malappuram and Wayanad districts after August 8.	https://m.cnr.cn/news/20190815/t20190815_524731977.html
3	2020	On August 7, 2020, continuous heavy rains led to a massive landslide in Rajamala, Idukki district, Kerala. The disaster occurred in the early morning hours, and many victims were unable to escape as they were asleep. By August 9, local police confirmed that the death toll had risen to 43, with around 20 people still missing.	http://zjnews.china.com.cn/yuanchuan/2020-08-10/242615.html
4	2021	By the evening of October 18, 2021, heavy rainfall and flash floods in Kerala had resulted in at least 23 fatalities due to landslides. The southern and central regions of Kerala, particularly the hilly areas of Kottayam and Idukki, were severely affected by the downpour, causing widespread landslides and flash floods. Preliminary reports indicated that 124 houses were completely destroyed in four districts, and 322 houses suffered partial damage.	https://news.sina.cn/gn/2021-10-19/detail-iktzqtyu2208227.d.html
5	2022	On August 29, 2022, a severe landslide occurred in the Maliyekkal area of Kudayathoor, near Thodupuzha in the Idukki district of Kerala, resulting in the unfortunate loss of at least five lives.	Amal George et al. (2023)
6	2024	On the early morning of July 30, 2024, due to the monsoon season, Wayanad district in Kerala experienced continuous heavy rainfall, which triggered large-scale landslides. The intense rainfall caused slope collapses, and debris flows, water, and boulders poured down the hillsides. This landslide resulted in at least 373 fatalities, over 200 injuries, and 218 people missing, marking it as one of the deadliest natural disasters in Kerala's history.	https://news.cctv.com/2024/08/05/ARTInm4iUwXBu5qoejp5AWdn240805.shtml

3. Landslide overview and precipitation conditions

Under the influence of the monsoon, the Wayanad region in Kerala, India, experienced continuous intense rainfall during the early morning hours of July 30, 2024 (02:00–04:00 IST), which triggered a large-scale landslide (Fig. 2). The continuous precipitation caused massive amounts of soil, rock, and vegetation to be swept downhill, resulting in significant destruction to roads, homes, and farmland throughout the area. The landslide led to at least 373 fatalities (Wu et al., 2025), over 200 injuries, and 218 individuals reported missing. Most of the victims were workers employed in tea and cardamom plantations, who were likely resting in temporary shelters when the landslide occurred. The majority of the casualties came from the regions of Mundakkai, Attamala, Chooralmala, and Kunhome in Wayanad.

The landslide originated approximately 8 km southwest of the Iruvanipuzha River, near a remote, uninhabited forest village, extending into surrounding tea plantations and forested areas. According to the Indian Meteorological Department (IMD) precipitation forecast, the northern part of Kerala, including Wayanad, was under an orange alert at the time, signaling that heavy to very heavy rainfall (ranging from 7 to 20 cm) was expected within a few hours. According to accounts from local residents, the first landslide occurred in the Chooralmala area at approximately 1:00 AM, followed by a series of successive landslides and debris flows in the Mundakkai region (Kolathayar et al., 2024). This cascading sequence of slope failures nearly devastated the entire village. Based on the landslide classification system proposed by Varnes (Hungr et al., 2014; Varnes, 1978), the event initially manifested as a planar sliding rock fall, which subsequently evolved into a debris flow. In addition, the collapse of the Punapuzha Bridge — the sole access route connecting Mundakkai to Chooralmala and other parts of Wayanad — resulted in the complete isolation of the Mundakkai area. This infrastructural failure severely hindered rescue and relief operations, posing significant challenges to emergency response efforts in the region.

Kerala, India, is characterized by a tropical monsoon climate, with abundant annual rainfall, particularly during the monsoon season, where the average annual precipitation is approximately 2400 mm. Specifically, the Wayanad region receives an average annual rainfall of 2786 mm. It is noteworthy that the southern, southwestern, and northeastern parts of Wayanad experience annual rainfall exceeding 3000 mm. Given that the landslide occurred during the monsoon season, rainfall intensity in Kerala was notably high at the time. On the day of the event, Kerala was undergoing intense rainfall. According to the Indian Meteorological Department, some areas of Kerala received exceptionally high rainfall during this period (Fig. 3), with local rainfall exceeding 200 mm. Specifically, the Thettamala rain gauge in Wayanad recorded 409 mm of rainfall over a 24-h period ending at 8:30 a.m. on July 30th, 2024 (Das, 2024). The heavy rainfall triggered catastrophic landslides and massive debris flows (Krishnapriya et al., 2024), which cascaded down along the Iruvanipuzha River at the source of the landslide. The continuous downpour caused significant amounts of soil, rock, and vegetation to slide down, devastating local roads, houses, and agricultural fields. The persistent heavy rainfall posed substantial challenges to rescue operations, as roads were slippery, visibility was low, and the progress of rescue efforts was significantly slowed.

4. Landslide causation analysis based on disaster system theory

The Western Ghats in Kerala have experienced sporadic landslide activities over the past few decades (Kuriakose et al., 2009; Lukose et al., 2010). However, since the catastrophic floods in 2018, landslide incidents have surged annually following episodes of extreme rainfall and flooding. This has led to a widely held academic consensus that extreme rainfall is the predominant driver of landslides in the region. While rainfall often serves as the primary trigger for landslides, it is rarely the sole controlling factor. For instance, despite the Nilambur region experiencing 398 mm of heavy rainfall on August 8, 2018 (Monsoon, 2018),

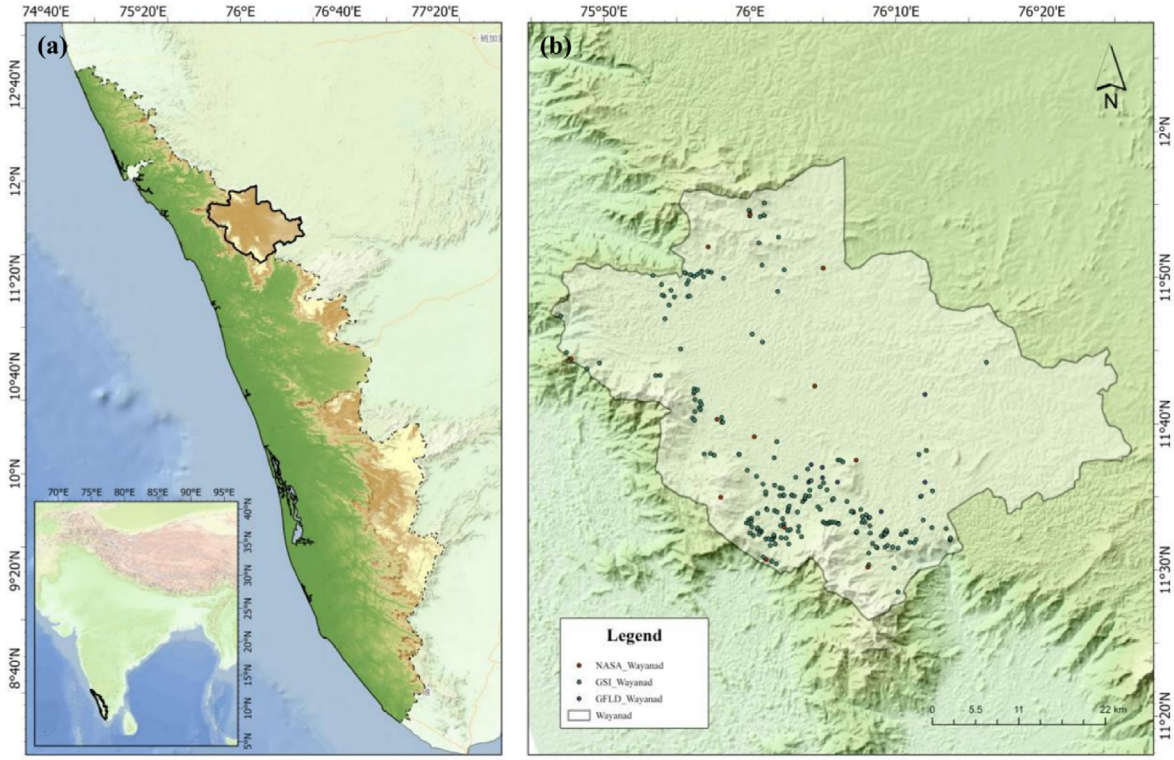

Fig. 1. (a) Location map of Wayanad region; (b) Distribution of historical landslide disasters.

Fig. 2. Large-scale landslide in Wayanad, Kerala, India.

no landslide occurred at Muthappankunnu. Therefore, attributing landslides solely to single-factor triggers is an oversimplification.

Building on this perspective, we employed disaster system theory (Fig. 4) to conduct a systematic causation analysis of the landslide event in Wayanad, India. Disaster system theory posits that disasters are the result of the complex interactions among the hazard-inducing environment, triggering factors, and exposed elements within the Earth's surface system (Li and Nie, 2016; Lu and Liao, 2024). Furthermore, the theory underscores that system vulnerability is the fundamental cause of disasters, with this vulnerability manifesting in the interrelations among exposed elements, triggering factors, and the hazard-inducing environment. Similarly, the Wayanad landslide event highlights the inherent vulnerability of the local system, where the interplay among these three components—exposed elements, triggering factors, and the hazard-inducing environment—emerged as the decisive factors contributing to the occurrence of this disaster.

4.1. Instability of the hazard-inducing environment

The rainfall-induced landslide event occurred in the southern part of Wayanad, a region characterized by hilly terrain with relatively steep slopes. Geological investigations reveal that the hill transects in this area are predominantly underlain by the Charnockite Group, which comprises narrow bands of pyroxene granulite and banded magnetite quartzite (Achu et al., 2025; Gopinath et al., 2024). This distinctive lithological and structural setting significantly increases the region's susceptibility to landslides under conditions of intense rainfall. Prolonged tropical weathering has resulted in the development of a thick, unconsolidated regolith at the surface, which becomes rapidly saturated during heavy precipitation, thereby leading to a drastic reduction in shear strength. Furthermore, the structural fabric of these rocks is prone to the development of weak zones, such as joints and foliation planes, along which failure is likely to initiate. In addition, certain rock layers exhibit low permeability, promoting the buildup of transient pore water pressure during rainwater infiltration, which further compromises slope stability.

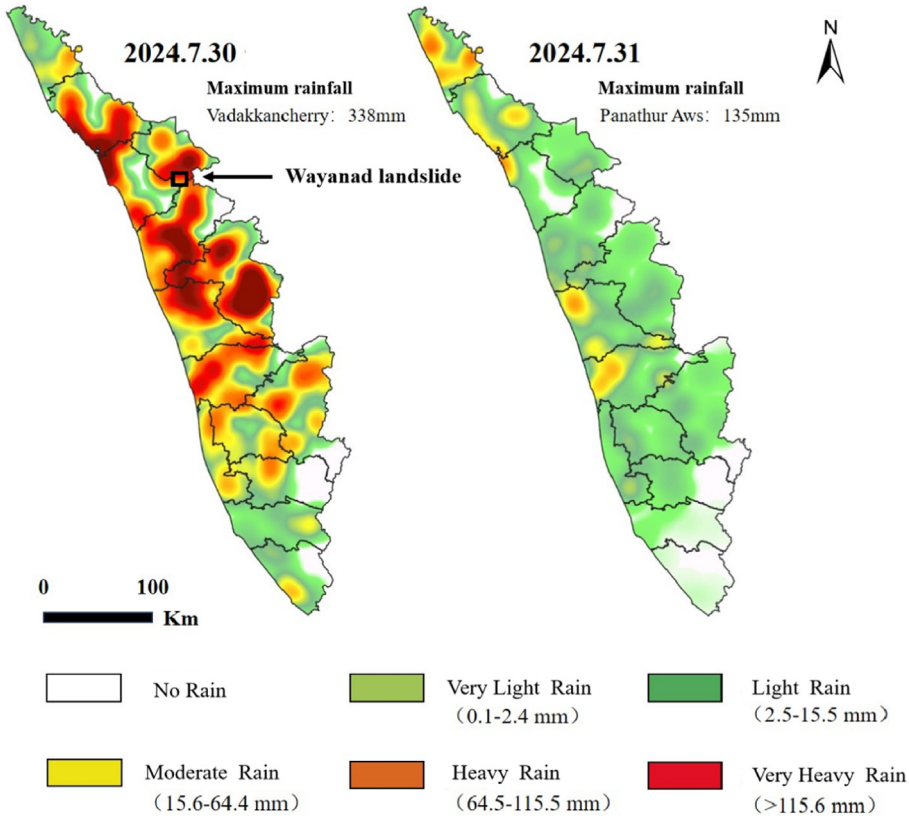

Fig. 3. Rainfall data for Kerala from July 30th to July 31st.
(image modified from (IMD, 2024)).

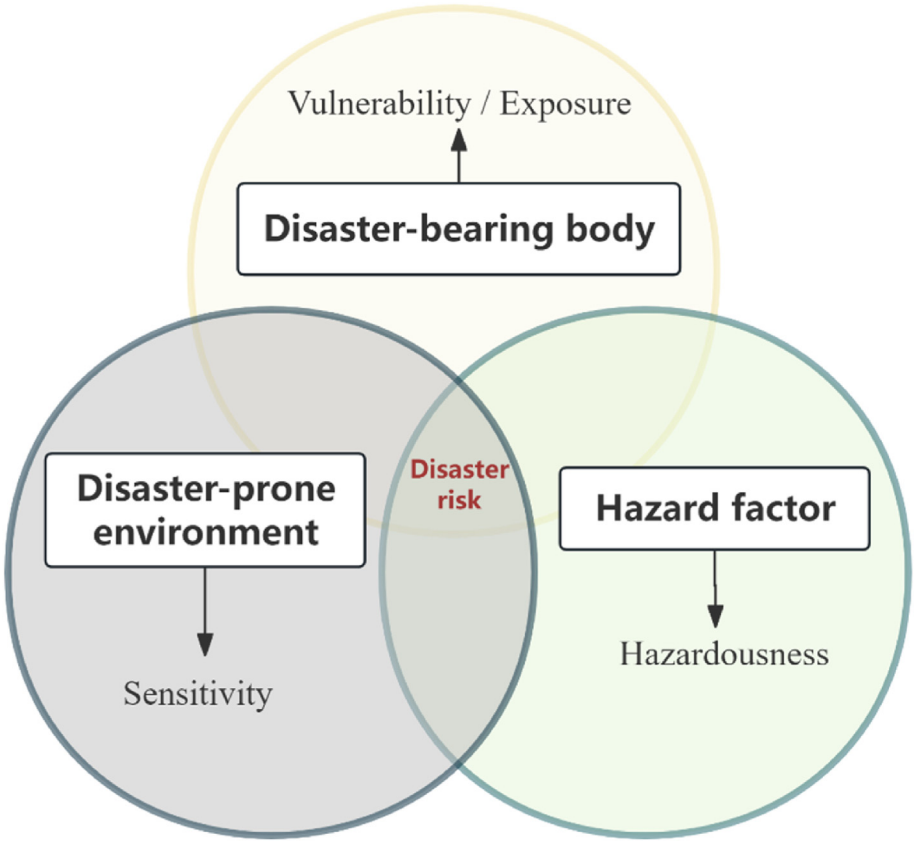

Fig. 4. The framework of the disaster system theory model.

The interplay of these factors renders the region particularly sensitive to shallow to intermediate-depth landslides under extreme rainfall events.

Moreover, a comparative analysis of pre- and post-event satellite imagery (Fig. 5) reveals that the landslide affected a broad area, with a wide runout path and substantial total impact zone. The remote sensing data also indicate that the 2024 landslide may represent a reactivation of a pre-existing landslide system. Historical imagery retrieved from the Google Earth platform shows clear signs of slope instability as early as the 2019–2020 monsoon season. In the subsequent years, the extent of slope failures appears to have progressively expanded, with minor landslides also observed along both sides of the adjacent stream channel (Fig. 6). Under such geomorphic and geological conditions, the region exhibits a high degree of susceptibility to reactivation under extreme rainfall. The earlier monsoon-induced landslide events may have structurally weakened the slope, leaving behind unconsolidated debris and residual failure surfaces that remained in a state of marginal stability. During the 2024 rainfall event, intense and concentrated infiltration of surface water significantly increased the pore water pressure within the regolith and weathered zones, further reducing the shear strength of the slope materials. This led to the reactivation or enlargement of the previous landslide body and adjacent unstable zones, ultimately resulting in the large-scale landslide disaster.

Meanwhile, in terms of land use and vegetation cover, the area surrounding the landslide source is densely vegetated, while the affected zone is primarily composed of tea plantations and mixed cropping areas, exhibiting a relatively high overall vegetation coverage. Between 1950 and 2018, the Wayanad region reportedly lost nearly 62 % of its forest cover, while the area under artificial plantations expanded by approximately 1800.75 % (Saha et al., 2022). Excessive deforestation, agricultural expansion, unregulated construction, and other human activities have disrupted the region's natural landscape, altered surface vegetation, and degraded soil structure. These changes have weakened natural barriers, rendering the soil more susceptible to erosion and increasing the risk of landslides. Additionally, irrational land use and a lack of effective soil and water conservation measures have exacerbated the landslide problem. As a prominent tourist destination in India, Wayanad has experienced significant growth in infrastructure development and related activities in recent years, which have, to some extent, compromised the natural stability of the slopes. Based on our analysis, these human activities are among the indirect factors contributing to the disaster, playing a major role in destabilizing slopes and causing landslides in the region.

4.2. Risk of triggering factors

4.2.1. High rainfall intensity

On July 30, the Wayanad region experienced exceptionally heavy rainfall, recording 572 mm within 48 h—far exceeding the anticipated 204 mm (https://www.reuters.com/business/environment/climate-change-intensified-rain-that-caused-deadly-indian-landslides-study-finds-2024-08-13/). This extreme intensity of rainfall rapidly increased pore water pressure within the soil, saturating the soil and rock masses in a short period, thereby compromising their stability and support. Moreover, the surface runoff generated by heavy rainfall intensified the erosion of slope material, further weakening the structural integrity of the slopes.

In addition to the immediate impacts of short-term heavy rainfall, prolonged rainfall in the weeks leading up to the disaster played a crucial role. Preceding rainfall significantly exceeded the long-term average, leaving the soil nearly saturated. The cumulative effect of prolonged rainfall elevated groundwater levels, breaching critical thresholds and triggering catastrophic large-scale landslides. Preliminary analysis indicates that the primary triggering factor for this catastrophic landslide in Kerala was the intense tropical rainfall associated with the monsoon season (Kuriakose et al., 2009; Achu et al., 2020; Feby et al., 2020; Wadhawan et al., 2020; GSI. et al., 2024).

4.2.2. Intensifying climate change

In recent years, the intensification of global climate change has led to a marked increase in the frequency of extreme weather events (Ranjan and Karmakar, 2024). The Wayanad landslide is, to some extent, a consequence of these long-term climate impacts. Research (Das, 2024) indicates that the rain belt along India's west coast has been shifting southward in recent years, altering the temporal and spatial distribution of rainfall. Concurrently, rising ocean temperatures in the Arabian Sea, driven by global warming, have provided the monsoon system with increased moisture. This has, in turn, intensified rainfall in regions like Wayanad, exacerbating localized precipitation intensity and concentration.

4.3. Vulnerability of the affected bodies

4.3.1. Inadequate disaster warning and emergency response capacity

Although India has strengthened meteorological monitoring and extreme weather warning systems in recent years, the capacity for monitoring and warning of landslides—localized and with complex triggering mechanisms—remains insufficient. In this disaster, while local meteorological and relevant authorities issued early warnings regarding heavy rainfall and landslides before the event, the notification time and the lead time for landslide forecasting were relatively short, leaving insufficient time for large-scale disaster preparedness efforts. For those living in high-risk areas, authorities failed to arrange timely evacuations, which exacerbated the severity of the disaster. Additionally, in remote and mountainous regions, poor communication further delayed residents

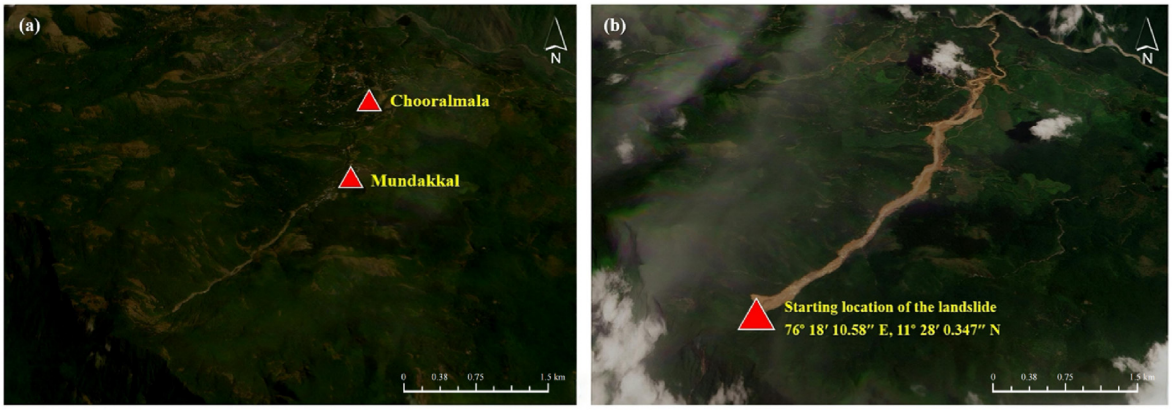

Fig. 5. Comparison of satellite remote sensing images before and after the landslide event. (a) May 2024; (b) right: August 2024. (All satellite imagery employed in this study is high-resolution, with a spatial resolution of 3 m)

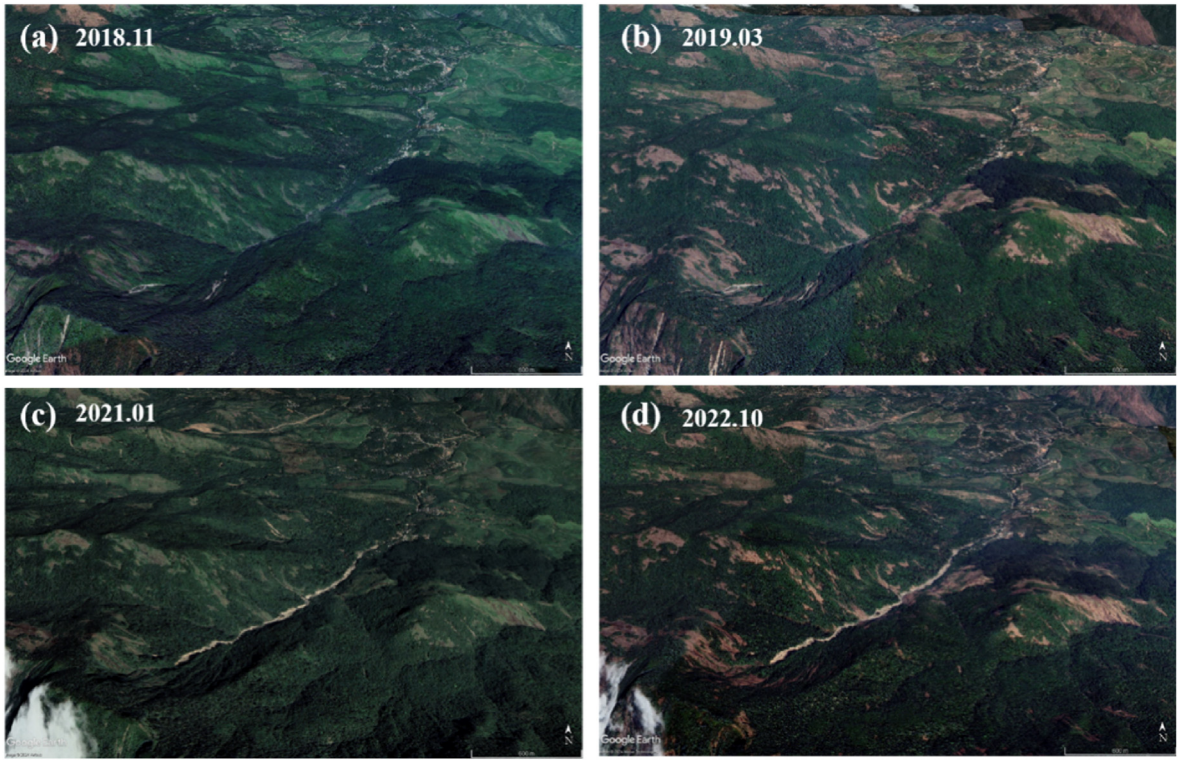

Fig. 6. Landslide extent changes in the region from 2018 to 2022.

from receiving timely warning information. Moreover, there has been a lack of systematic landslide risk assessments in related planning and construction efforts. Some infrastructure projects (such as roads, drainage systems, and tourism developments) did not fully consider geological conditions and even directly undermined slope stability. The absence of scientific planning and effective governance measures has further amplified the disaster risk. In terms of emergency response, a collaborative model involving data sharing, information exchange, and joint participation has not yet been established among various departments. Consequently, rescue equipment and personnel had difficulty reaching the disaster site swiftly and accurately after the landslide, delaying critical rescue operations.

4.3.2. Lack of public awareness and preparedness for disaster risk

With advancements in communication technologies and comprehensive satellite monitoring, relevant authorities now issue warnings about heavy rainfall through multiple channels in advance. However, the population in many of these areas mainly consists of workers in tea and cardamom plantations, most of whom have limited educational backgrounds and lack disaster awareness. This results in indifference to the early warning notifications. Additionally, some regions experience delays or inadequate dissemination of warning messages, preventing timely evacuation before the disaster. This led to significant loss of life and property, as people could not evacuate quickly enough.

4.3.3. High social risk

The Wayanad region, as a major ecological tourism hotspot and agricultural area in southern India, has a high population density, especially in villages and farmlands near slopes. In response to population growth and economic development, many residents have cultivated agricultural land or built homes on geologically unstable slopes. This has resulted in irrational land use practices, which have weakened the natural stability of the slopes and increased the community's exposure to disaster risks. Furthermore, many local residents rely on agriculture for their livelihood and have relatively low incomes. Low-income groups tend to live in high-risk areas with poor geological conditions, where housing structures are generally not sturdy enough, and disaster protection measures are lacking. After the landslide, the life and property losses for this group were especially severe, and their limited economic resources and recovery capacity significantly hindered post-disaster recovery, further exacerbating overall social risk. Additionally, inadequate drainage systems in some areas worsened flooding due to heavy rainfall, further increasing the risk of landslides. These factors, when combined, significantly elevated the overall social risk in the Wayanad region.

4.4. Comparison of disaster system theory with other methods

Based on the Disaster System Theory, we conducted a preliminary analysis of the causal mechanisms underlying the extreme landslide event that occurred in Wayanad, India, in 2024. Our analytical results are highly consistent with several key arguments presented in previous research. For instance, Petley (2012) and Kirschbaum et al., 2010, 2015 have argued that the increasing landslide risk across South Asia is primarily driven by the compounded effects of unstable geological structures, intensified human interventions, and extreme weather conditions—factors that are particularly pronounced in the Wayanad region. Similarly, Jain et al. (2021) emphasized that landslide events in Kerala are typically the result of the confluence of extreme rainfall, geological instability, and anthropogenic pressures. Wadhawan et al. (2020), Kuriakose et al. (2009), and Ajin (Ajin et al., 2022) further highlighted that the frequent landslides in Kerala and specifically in Wayanad cannot be attributed to rainfall alone; rather, they arise from a systemic coupling of climatic triggers, human activities, insufficient early warning mechanisms, environmental degradation, and land-use mismanagement.

Our analysis of the 2024 Wayanad landslide aligns with this systemic framework: the event resulted from the synergistic interaction of a hazard-prone environment, triggering factors, and vulnerability of the exposed elements. More specifically, Wayanad's hazard-prone environment is characterized by highly weathered bedrock and steep terrain

gradients, which together form a natural precondition for slope failure. The primary triggering factor was an episode of high-intensity rainfall under the broader influence of climate change, which significantly increased soil saturation and initiated slope instability. The vulnerability of the disaster-affected population was magnified by the concentration of human settlements in susceptible zones, overdevelopment of land in hazard-prone areas, and insufficient early warning and emergency response mechanisms. Fraser-Baxter (2024) and Das (2024), in their independent analyses of the 2024 event, similarly concluded that the landslide was the result of a complex interplay between geological, geomorphological, and climatic factors. In addition to these established elements, our study introduces a relatively underexplored perspective in Indian landslide literature: the relationship between plantation-driven deforestation and systemic vulnerability. Wayanad has undergone prolonged plantation expansion, leading to a marked reduction in forest cover. This decline has weakened the root-reinforcement effect of vegetation on soil cohesion, thereby reducing slope stability. Furthermore, plantation development is often accompanied by extensive slope modification and the construction of artificial drainage systems, which have further disrupted the geotechnical balance of the region. Under conditions of extreme rainfall, these anthropogenic interventions significantly exacerbate slope failure potential. Therefore, the disaster system theory not only has good theoretical adaptability, a comprehensive perspective, and sufficient practical explanatory power in analyzing such complex geological disasters, but it can also provide new research directions and practical insights for future disaster prevention and mitigation measures.

5. Risk prevention and emergency response measures

5.1. Pre-disaster phase: monitoring, early warning, and risk prevention

India has established a multi-tiered geological hazard risk monitoring and early warning system at the pre-disaster stage, with the state of Kerala being no exception. In the domain of landslide forecasting and early warning, a significant milestone was marked on July 19, 2024, when the Geological Survey of India (GSI) operationalized the National Landslide Forecasting Centre (NLFC). The system was initially designed to provide daily forecasts for landslide-prone regions such as Kalimpong, Darjeeling, and the Nilgiris, with plans to extend its coverage to other vulnerable areas nationwide by 2030 (https://believersias.com/national-landslide-forecasting-centre/). In addition, the NLFC has been providing experimental forecast data to 13 other landslide-prone areas, including Kerala (https://bhusanket.gsi.gov.in/Activities_V9.html); however, during the event in question, such information was disseminated solely to state and district-level disaster management authorities rather than directly to the general public (https://www.hindustantimes.com/india-news/wayanad-tragedy-how-kerala-can-improve-its-early-warning-systems-for-landslides-101722510782490.html). This lack of direct public dissemination significantly constrained residents' capacity for timely response. Moreover, a local research institution in Wayanad issued a landslide risk alert based on observed rainfall approximately 16 h prior to the disaster (https://www.thenewsminute.com/kerala/wayanad-landslide-environmentalists-say-mundakkai-region-uninhabitable-warnings-ignored). However, as this warning was not incorporated into the official early warning system, it was not acknowledged by the administrative authorities. There remains considerable contention regarding the accuracy and interpretability of the official forecasts prior to the disaster. The Chief Minister of Kerala, Pinarayi Vijayan, stated that on July 29 and 30, the central government's warning system had classified the landslide risk for Wayanad as "low probability." Similarly, the Central Water Commission (CWC) did not issue any flood alerts for the Iruvazhinji or Chaliyar rivers between July 23 and 29 (https://www.onmanorama.com/news/kerala/2024/07/31/wayanad-landslides-kerala-cm-pinarayi-vijayan-update.html). In contrast, the central government contended that an Orange Alert for heavy rainfall had been issued for July 29, which included the possibility of landslides and localized flooding. Additionally, the GSI issued a landslide warning for Wayanad at 2:00 p.m. on July 29, arguing that the state authorities had "adequate lead time" to initiate precautionary measures (https://www.newindianexpress.com/states/kerala/2024/Aug/03/reliable-rainfall-data-must-for-accurate-landslide-warning-gsi).

In terms of meteorological forecasting, the India Meteorological Department (IMD) released relevant weather alerts prior to the landslide event. According to publicly available records on Wikipedia, the central government issued four early warnings to the Kerala government between July 23 and 26, specifically highlighting the likelihood of intense rainfall exceeding 200 mm in the Wayanad region (https://en.wikipedia.org/wiki/2024_Wayanad_landslides#Background). On July 29, the IMD issued an Orange Alert for Wayanad, forecasting rainfall in the range of 115.6 mm–204.4 mm. However, the actual precipitation amounted to 372 mm (https://www.thehindu.com/news/national/kerala/wayanad-landslide-how-two-villages-vanished-overnight/article68506601.ece), substantially surpassing projections and significantly escalating the scale of the landslide. Nonetheless, the meteorological warnings have also been subject to controversy. The Chief Minister emphasized that the highest level of alert—a Red Alert indicating an extremely high risk of torrential rainfall—was issued only after the landslide had already occurred (on July 30 at 6:00 a.m.) (https://www.onmanorama.com/news/kerala/2024/07/31/wayanad-landslides-kerala-cm-pinarayi-vijayan-update.html), while central agencies maintain that sufficient early warnings were indeed disseminated in advance (https://www.indiatoday.in/india/kerala/story/wayanad-landslides-centrel-governmnet-to-pinarayi-vijayan-claims-on-early-warning-2576048-2024-08-02).

Regarding geological survey and risk assessment, the Geological Survey of India had previously undertaken landslide susceptibility mapping across the Western Ghats, categorizing the affected region of Wayanad as a high-sensitivity zone. A pilot AI-based landslide early warning system had also been deployed in Wayanad approximately ten days prior to the event. However, as the system was still under calibration and optimization, it failed to predict the landslide accurately (https://www.newindianexpress.com/states/kerala/2024/Jul/31/early-warning-system-for-natural-disaster-fails-wayanad). According to GSI, the predictive accuracy of this early warning system is heavily dependent on rainfall data provided by the IMD, which in July 2024 was insufficient and lacked precision. As a result, the system was unable to generate accurate landslide warnings for Wayanad (https://www.newindianexpress.com/states/kerala/2024/Aug/03/reliable-rainfall-data-must-for-accurate-landslide-warning-gsi).

In the realm of public disaster preparedness education, local governments in Kerala have implemented several initiatives aimed at enhancing community resilience. Certain villages—Mundakkai, Puthumala, Attamala, Vellithodu, Thrikkaippatta, and Kottatharavayal—have been designated as landslide-prone zones, thereby elevating public awareness in those areas (https://www.onmanorama.com/news/kerala/2024/08/21/wayanad-landslides-master-plan-swept-away-nature-fury-chooralmala-meppadi.html). In 2023, the Kerala State Disaster Management Authority (KSDMA) conducted school safety training programs (July) and activities commemorating International Day for Disaster Risk Reduction (October 13) in Wayanad, targeting school children and local communities (https://sdma.kerala.gov.in/events-2023/). Additionally, the KSDMA developed and distributed educational brochures, such as the Landslide Safety Guide, to improve community knowledge and readiness (https://sdma.kerala.gov.in/brochures/). Despite these efforts, the actual reach and effectiveness of these programs may have been limited, especially in the face of an event whose magnitude exceeded prior expectations. For instance, on July 29, due to torrential rainfall, local authorities began relocating residents from Punjirimattom, Attamala, and Mundakkai to relief camps. However, a significant number of residents in Chooralmala chose to remain, as the area had not been formally classified as a high-risk zone. This reflects lingering gaps in public awareness and risk perception regarding disaster preparedness (https://en.wikip

edia.org/wiki/2024_Wayanad_landslides).

5.2. During-disaster phase: emergency response and rescue operations

Following the occurrence of the landslide, a multi-tiered emergency response mechanism was promptly activated by authorities at all administrative levels, initiating a coordinated, multi-agency rescue operation. Composite rescue teams comprising personnel from the National Disaster Response Force (NDRF), the Indian Navy, and the state fire and rescue services were rapidly deployed to the affected site, undertaking nighttime rescue operations and the evacuation of survivors (https://timesofindia.indiatimes.com/india/wayanad-landslides-rescue-ops-stepped-up-as-several-remain-missing/photostory/112176014.cms?picid=112176034). The Kerala state government immediately elevated its disaster response level, while the central government provided significant logistical and operational support. Specialized NDRF units were airlifted from adjacent states and deployed to the disaster zone with minimal delay. Furthermore, the Indian Army Corps of Engineers and the Indian Air Force contributed to the search-and-rescue (SAR) operations and aerial delivery of essential relief supplies. Given the destruction of key transport infrastructure—including critical bridges leading to epicentral villages—the Indian Army constructed a 58-m-long temporary steel bridge across the Iruvanipzha River, thereby restoring access and enabling the timely movement of heavy machinery and relief vehicles (https://www.indiatoday.in/india/kerala/story/wayanad-landslides-indian-army-bridge-in-record-time-chooralmala-mundakkai-iruvanipzha-river-2575346-2024-08-01). Military and paramilitary rescue personnel employed techniques such as rope-assisted crossings and aerial zipline systems to evacuate stranded survivors across swollen rivers. Within the first 72 h post-disaster, several individuals trapped beneath the debris were successfully rescued alive. Advanced technologies, including unmanned aerial vehicles (UAVs), were deployed to support search operations using thermal imaging systems, which facilitated the localization of human remains buried under deep debris flows (https://www.indiatoday.in/india/story/wayanad-kerala-landslides-death-toll-survivors-rescue-operations-rahul-priyanka-gandhi-pinarayi-vijayan-2575468-2024-08-02). In addition to state-led efforts, extensive community participation played a pivotal role in the emergency response. Local civil society groups, neighborhood volunteer networks, and non-governmental organizations mobilized promptly, contributing to both rescue operations and relief distribution (https://www.thehindu.com/news/national/kerala/selfless-service-by-volunteers-of-civil-groups-in-relief-operations-in-landslide-hit-wayanad/article68516427.ece). On the medical front, the Kerala Department of Health activated an emergency medical coordination centre in the immediate aftermath of the landslide. Medical reinforcements were mobilized from surrounding hospitals, while frontline health personnel were deployed to Wayanad to augment capacity. All regional ambulance stations were stocked with essential medications and equipped with sufficient emergency beds to treat a large influx of injured individuals.

As a result of these multi-sectoral and multi-scalar collaborative efforts, approximately 4000 displaced persons were evacuated and temporarily resettled within the first week following the disaster, housed across 45 emergency relief shelters established by the authorities (https://www.indiatoday.in/diu/story/kerala-wayanad-landslide-crisis-monsoon-fury-rescue-operation-natural-disasters-himachal-uttarakhand-cloudburst-2574826-2024-08-01).

This comprehensive emergency operation exemplifies the effective functioning of India's federal disaster management architecture, which is characterized by vertical coordination among central, state, and local institutions and robust civil–military integration. The central government provided critical personnel and logistical resources, the state government assumed operational command, and local governments—alongside civil society actors—implemented ground-level response and rehabilitation measures. Collectively, this multi-level synergy substantially mitigated the extent of secondary casualties and ensured rapid relief delivery in a highly complex terrain.

5.3. Post-disaster phase: recovery, reconstruction, and risk management

After the conclusion of emergency search-and-rescue operations, both the Government of India and the Government of Kerala initiated a phased recovery and reconstruction program, combining short-term relief efforts with long-term disaster risk reduction (DRR) strategies.

In the immediate aftermath of the disaster, priority was given to the resettlement of displaced populations and the distribution of emergency relief supplies. The Prime Minister of India announced ex gratia compensation from the Prime Minister's National Relief Fund, providing ₹200,000 to the families of the deceased and ₹50,000 to injured individuals (https://reliefweb.int/report/india/situation-report-2-wayanad-landslide-30th-july-2024). Concurrently, the Kerala state government activated the Chief Minister's Distress Relief Fund to mobilize public donations and issued interim subsistence allowances for families who had lost their homes (https://www.newindianexpress.com/states/kerala/2024/Aug/03/kerala-government-plans-new-safe-township-for-rehabilitation-of-landslide-victims-in-wayanad). Public health authorities implemented intensified epidemic prevention and vector control measures, including the disinfection of affected environments and removal of decomposing matter, in order to prevent the outbreak of post-disaster communicable diseases. Technical agencies and geoscientific experts were promptly dispatched to conduct on-site hazard evaluations (https://www.deccanchronicle.com/southern-states/kerala/five-member-expert-team-to-visit-landslide-hit-wayanad-today-1815996). Emergency geotechnical assessments conducted by the Geological Survey of India (GSI) and other institutions focused on the stability of landslide debris and adjacent slopes (https://www.onmanorama.com/content/mm/en/kerala/top-news/2024/08/10/wayanad-landslide-velocity-rated-extremely-rapid-covered-seven-km-says-gsi-report.html). Stabilization measures were carried out to secure potentially loosened rock and soil masses. Hydrological interventions included the clearance and dredging of the Chaliyar River, which had been obstructed and diverted by debris flows, in order to restore normal fluvial discharge and mitigate the risk of secondary flooding (https://www.thehindu.com/news/national/kerala/wayanad-landslides-chaliyar-rivers-serene-waters-turned-deadly/article68484186.ece). In the later stages of SAR operations, large-scale deployment of labor and heavy machinery facilitated the removal of several million cubic meters of landslide debris by mid-August. A dedicated panel of experts was convened to assess the post-disaster habitability of the affected region. Based on field surveys, certain sectors within the three most severely impacted villages were formally designated as permanent high-risk zones, thereby prohibiting reconstruction and long-term habitation in those areas (https://www.newindianexpress.com/states/kerala/2025/Jan/02/rs-750-crore-township-project-unveiled-for-rehabilitation-of-wayanad-landslides-victims).

From a strategic resilience perspective, the state government adopted a policy of planned relocation, opting to construct new resettlement townships in safer locations. Within 61 days of the disaster, the Kerala government approved the repurposing of two abandoned tea estates in Wayanad district for the establishment of new residential townships under a comprehensive post-disaster reconstruction plan (https://www.newindianexpress.com/states/kerala/2025/Jan/02/rs-750-crore-township-project-unveiled-for-rehabilitation-of-wayanad-landslides-victims). In addition to housing reconstruction, livelihood restoration constituted a key pillar of the long-term recovery framework. The local administration implemented a “Micro-Rehabilitation Scheme” to provide financial and infrastructural support for micro, small, and medium-sized enterprises (MSMEs), thereby fostering economic revitalization. Simultaneously, infrastructure rehabilitation efforts were launched, including the construction of replacement bridges and roads to reestablish transportation connectivity. Strategic investments were made to develop auxiliary emergency bridge networks to enhance transport redundancy and resilience against future extreme events (https

://www.newindianexpress.com/states/kerala/2024/Aug/03/kerala-government-plans-new-safe-township-for-rehabilitation-of-landslide-victims-in-wayanad). Parallel to physical reconstruction, the government embarked on institutional reform aimed at enhancing early warning and governance mechanisms. In January 2025, the state launched the Kavacham Integrated Disaster Warning System (Kerala Warnings, Crisis and Hazards Management), developed by the Kerala State Disaster Management Authority (KSDMA) with support from the World Bank and the National Disaster Management Authority (NDMA). This system consolidates all meteorological, hydrological, and geotechnical warning streams into a unified platform operated by the State Emergency Operations Centre. It includes the installation of 126 alarm and beacon towers in suburban and high-risk zones, complemented by 93 emergency response centers equipped with decision-support systems. Warnings are disseminated through multiple channels, including radio, SMS, and social media platforms, to ensure rapid community-level awareness (https://x.com/_ApniPathshala/status/1882687466362081409). A notable feature of the new system is its emphasis on enhancing monitoring and early warning for mountain hazards. Dense rain gauge networks and landslide sensor arrays are planned for deployment across high-risk zones of the Western Ghats. Institutional coordination is being strengthened through data-sharing protocols with IMD, GSI, and related agencies to address previously fragmented warning systems. Furthermore, the Wayanad landslide has catalyzed broader policy introspection on ecological management and regulatory oversight in the Western Ghats. Eminent ecologist Madhav Gadgil had long warned of the region's ecological fragility and called for curbs on anthropogenic pressures such as mining and plantation expansion (https://frontline.thehindu.com/environment/climate-change-environmental-disaster-landslides-in-india-kerala-wayanad-landslide-irshalwadi-landslide/article68473788.ece). In response, the Kerala government announced its intention to intensify regulatory enforcement in ecologically sensitive zones. Efforts are now underway to implement the Gadgil Report's zoning-based land-use controls, including stricter enforcement of construction codes and slope-use regulations to prevent hazardous development on geologically unstable terrains.

6. China's response experience and lessons from the wayanad landslide disaster

6.1. China's response experience and lessons for India

China has accumulated systematic governance experience in managing rainfall-induced landslide disasters. Since the beginning of the 21st century, significant progress has been made in various aspects, including geological hazard investigation and assessment, monitoring and early warning, integrated prevention and control, emergency response, and information technology, leading to remarkable results in disaster risk reduction. (1) Establishment of a comprehensive monitoring and early warning system. China has developed a nationally unified geological hazard monitoring and forecasting network, supported by a multi-agency early warning dissemination mechanism (Zhang, 2020). This system horizontally connects 16 key departments, including meteorology and natural resources, and vertically integrates the national, provincial, municipal, and county levels, enabling early warnings to be sent directly to grassroots-level officials (https://www.cma.gov.cn/en/special/20240618/2024061802/202411/t20241105_6675812.html). When high-level warnings are issued, meteorological authorities immediately notify township leaders and village-level disaster prevention personnel in the affected areas, instructing them to initiate hazard inspections and organize evacuations. This response mechanism has proven effective in landslide-prone regions such as the Yunnan–Guizhou Plateau (https://www.cma.gov.cn/en/special/20240618/2024061804/202302/t20230207_5292734.html) (Fan et al., 2019). (2) Comprehensive geological hazard surveys and risk assessments. The government mobilizes geological experts to conduct grid-based investigations and classification of landslide-prone sites, and mandates geological hazard risk assessments for major infrastructure projects, thereby preventing construction-induced landslide risks at the source (https://m.cgs.gov.cn/cgkx/202108/t20210825_680321.html). For villages with severe hazard exposure, China adopts a combination of engineering interventions and relocation strategies—such as slope stabilization, construction of retaining walls and drainage systems, and the full relocation of residents in extremely high-risk areas—fundamentally reducing population exposure (https://www.gov.cn/xinwen/2023-01/10/content_5736065.htm). (3) Strengthens public awareness campaigns and community-based disaster monitoring networks. Public outreach is enhanced through activities such as National Disaster Prevention and Reduction Day, community-based emergency drills, and science popularization events (http://www.rmlt.com.cn/2025/0319/725714.shtml). Locally established monitoring networks, with participation from residents, enable timely reporting of rainfall intensity and potential hazards, embodying the "mass monitoring and mass prevention" approach (Wu et al., 2020). (4) Development of an efficient emergency response and rescue mechanism. In the event of a landslide, China activates a unified emergency coordination mechanism led by the Emergency Management Department, which mobilizes firefighters, armed police units, geological and meteorological experts to the site. A multi-agency disaster command system is quickly established to ensure timely search, rescue, and relief efforts within the critical response window (https://www.mem.gov.cn/gk/zcjd/201901/t20190122_231712.shtml). Post-disaster, equal emphasis is placed on reconstruction and long-term mitigation—ensuring proper resettlement of affected populations, and implementing ecological restoration and engineering reinforcements of landslide slopes to enhance the region's resilience to future events.

In comparison, the Wayanad landslide in India exposed several shortcomings in disaster governance. For instance, although multiple warnings about heavy rainfall and potential landslides were issued by meteorological authorities, local governments failed to organize timely evacuations. The affected area had long suffered from deforestation, illegal land reclamation, and unregulated development for plantations and tourism, all of which weakened slope stability and exacerbated landslide risks. Alarmingly, as early as 2020, several villages engulfed in the 2024 disaster had already been identified as high-sensitivity landslide risk zones in official hazard assessment reports. However, the Indian government had not implemented adequate engineering controls or evacuation measures prior to the event, and hazard recognition failed to translate into effective risk management.

To address these gaps, India could draw on China's successful practices to improve its landslide disaster prevention and control framework. At the policy level, strict ecological protection and land use control regulations should be enacted, particularly targeting the landslide-prone areas of the Western Ghats (Deepu et al., 2020; Yunus et al., 2021). Fragile areas should be designated as ecologically sensitive zones with restricted development. Risk-prone villages should undergo planned relocation to reduce population exposure at the source (Singh and Parkash, 2019). At the technical level, India should build a real-time monitoring and early warning platform for rainfall-induced landslides in mountainous regions (Ramesh et al., 2023a). By integrating meteorological thresholds, remote sensing data, and IoT-based sensor networks, authorities could conduct continuous slope stability monitoring. In cases of abnormal deformation or extreme rainfall forecasts, timely alerts should be issued at the grassroots level, triggering mandatory evacuation protocols to ensure that early warnings result in concrete action. Institutionally, a warning responsibility system should be established, similar to China's model, clarifying the roles and accountability of government actors at all levels during the warning-to-response chain. This ensures that early warnings are not ignored, and responsibilities are fulfilled down to the "last mile." Public awareness and self-rescue capabilities must be strengthened. Grassroots monitoring teams formed by local residents (Matpady et al., 2023; Bhatia, 2013; Ramesh et al., 2023b)—modeled after China's "mass monitoring and prevention"

approach—can play a key role in identifying early signs of danger and reporting them promptly. Through coordinated improvements in policy, technology, and institutional mechanisms, India can gradually enhance its ability to prevent and manage landslide disasters, thereby minimizing the recurrence of such tragic events to the greatest extent possible.

6.2. Lessons from the wayanad disaster for China's landslide disaster prevention and response

The catastrophic landslide disaster in Wayanad, India – triggered by unprecedented heavy rainfall that caused massive slope failures – provides sobering lessons and insights for China's own landslide disaster prevention and response efforts. This event underscores that even countries with advanced systems must continuously evaluate and improve their disaster preparedness, especially in the face of evolving environmental conditions. Several key lessons from the Wayanad landslide can inform China's future strategies to ensure it remains robust against similar extreme events.

First, the Wayanad disaster highlights the crucial importance of continuously refining early warning systems and risk communication. In Wayanad, intense rainfall in a short period overwhelmed the usual warning thresholds and caught many residents off guard. For China, which already operates sophisticated landslide early warning models, this tragedy is a reminder that these models must be frequently updated to account for the increasing variability of climate patterns. Extreme precipitation events that were once rare are becoming more common; therefore, Chinese authorities should incorporate greater safety margins and real-time adaptability into their warning criteria. Equally important is ensuring that warnings effectively reach all at-risk communities, including those in remote mountainous areas. The Wayanad case showed that if communication networks fail or if warnings do not prompt immediate action, the best forecasting technology cannot save lives. China must continue to improve how warnings are disseminated at the local level — for instance, through multilingual alerts, community loudspeaker systems, and education that builds trust in official warnings — so that when alarms are raised, people respond rapidly and appropriately.

Second, the events in Wayanad illustrate the critical need to address underlying environmental and land-use factors as part of disaster prevention. Investigations into the Wayanad landslide revealed that years of deforestation, expansion of plantations, and construction on steep slopes had exacerbated the terrain's vulnerability. The heavy rains simply pushed an already unstable environment past its tipping point. This serves as a cautionary lesson for China: alongside technological and engineering solutions, strict environmental stewardship is vital for reducing landslide risk. China has many regions with fragile ecosystems (such as in the southwestern provinces and the Himalaya-adjacent areas), and development pressures can lead to slope destabilization if not managed carefully. The Wayanad tragedy reinforces the importance of enforcing land-use regulations and promoting sustainable development practices in such areas. Chinese authorities should continue to prohibit excessive deforestation and reckless hillside excavation, and instead encourage reforestation and soil conservation programs in landslide-prone localities. By maintaining healthy vegetation cover and managing water runoff (for example, through terracing and building proper drainage infrastructure), China can strengthen the natural resilience of slopes. In essence, the key lesson is that disaster prevention must begin long before rainfall starts – it begins with how we treat our environment and plan our communities on vulnerable terrain.

Moreover, the Wayanad disaster underscores that community preparedness and public awareness are indispensable components of effective disaster management. Many victims in Wayanad were caught in their homes at night, possibly unaware of the imminent danger or unsure of how to respond. This tragic outcome emphasizes that even with good monitoring, an uninformed or unprepared population will remain at great risk. China can learn from this by redoubling its efforts to educate and involve local communities in disaster readiness. In practical terms, this means conducting regular community-based drills for landslide evacuation, training residents to recognize early warning signs like unusual sounds or ground movement, and establishing local volunteer networks to assist in rapid response. It also involves culturally sensitive education campaigns to ensure that warnings are taken seriously – for instance, dispelling any false sense of security that "it won't happen here." When citizens are well-informed and trained, they are far more likely to evacuate promptly or take other lifesaving actions during the narrow window between a warning and a landslide. The lesson from Wayanad is clear: technology and infrastructure must be complemented by a populace that knows how to react when nature strikes.

Finally, the Wayanad landslide serves as a stark reminder of the ever-growing challenges posed by climate change and the necessity of adaptive, forward-looking disaster management. Climate scientists warn that as global temperatures rise, the frequency of extreme weather events – including intense monsoon downpours – is likely to increase. Disasters like the one in Wayanad, which saw an extraordinarily high rainfall within 48 h, could foretell the kind of scenarios that even China might face more often in the future. Therefore, China must incorporate climate change projections into its disaster planning. This entails using the latest climate models to identify regions where the risk of severe landslides is escalating and developing contingency plans accordingly. It also means fostering a mindset of continuous learning: each disaster, whether at home or abroad, provides data and insights to refine existing policies. By studying events like Wayanad in detail – understanding what went wrong and what measures were effective – Chinese experts can uncover potential blind spots in their own systems. This willingness to learn from others' experiences is a hallmark of mature disaster management. Going forward, China should strengthen international collaboration in disaster risk reduction, sharing information and strategies with countries like India, so that both can mutually improve their resilience. In conclusion, the Wayanad tragedy, while devastating, has yielded important lessons that can catalyze improvements in China's landslide disaster preparedness. By heeding these lessons – enhancing early warnings, protecting the environment, empowering communities, and anticipating future climate-driven extremes – China will be better equipped to prevent and respond to landslide disasters, thereby safeguarding lives and livelihoods in the years to come.

CRediT authorship contribution statement

Qinxia Wang: Writing – original draft, Formal analysis. **Chong Xu:** Writing – review & editing, Funding acquisition, Conceptualization. **Junxue Ma:** Formal analysis. **Yuandong Huang:** Formal analysis. **Shuhui Zhang:** Conceptualization. **Huiran Gao:** Data curation.

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Declaration of competing interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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